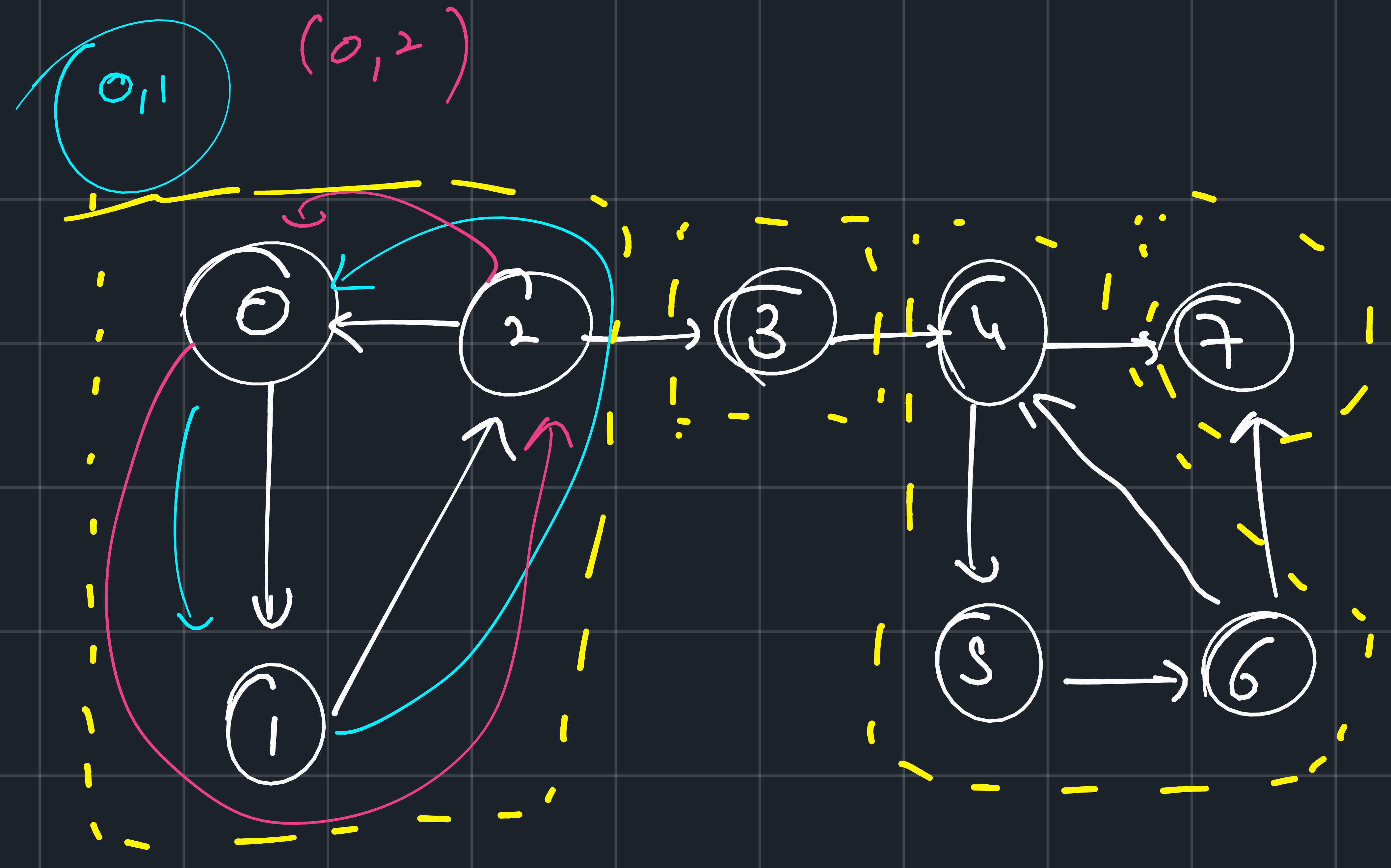


(0,1,2) (3) (4,5,6) (7)

↳ Kosaraju's Algorithm

↳ Helps us to find

↳ no. of strongly connected components. → only valid in directed cyclic graph.

↳ print the strongly connected component.

→ strongly connected components are

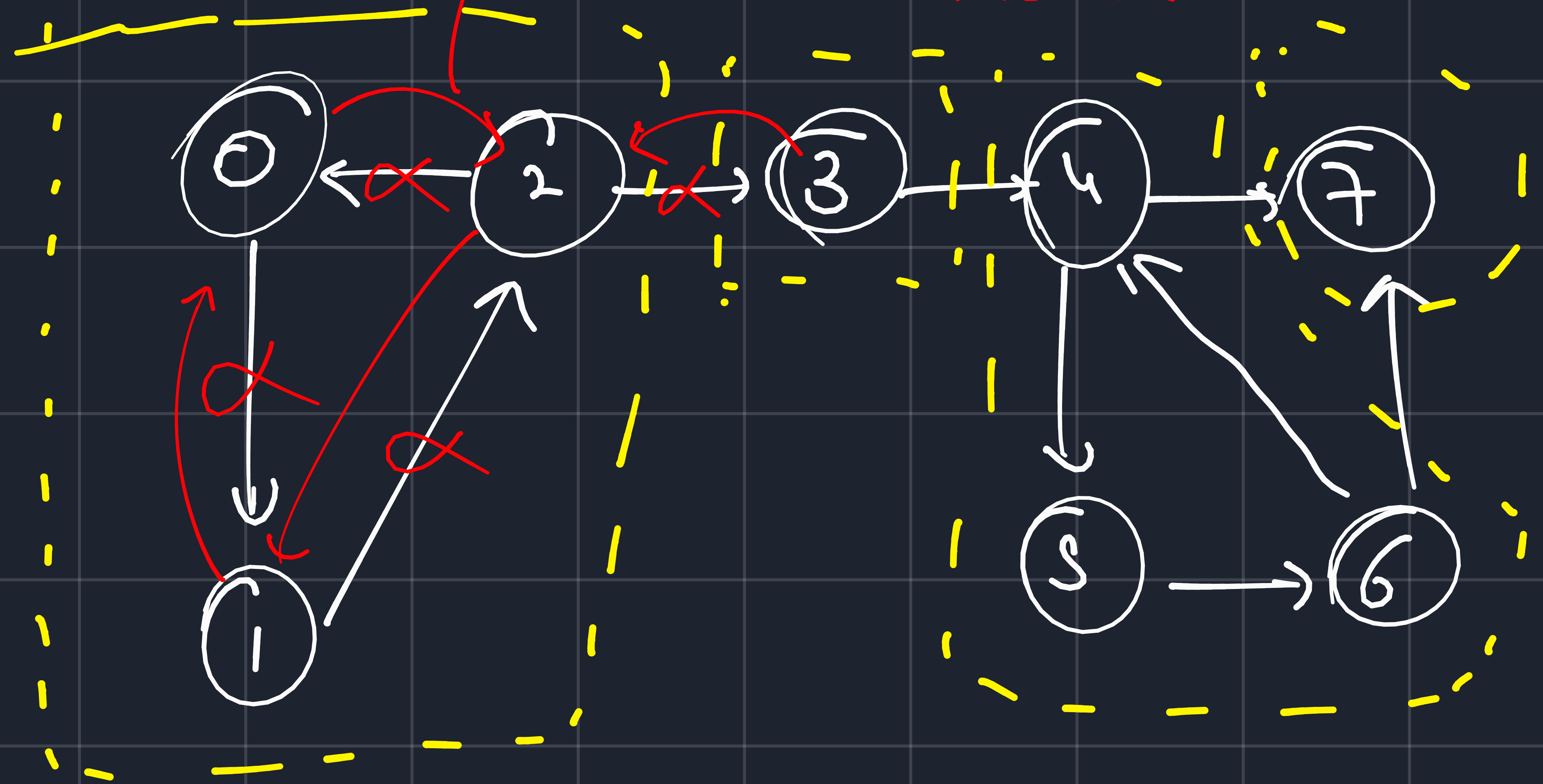
take any pair of vertices

u, v

u ⟶ v

v ⟶ u

Reversing the edges in a s.c.c does not matter.

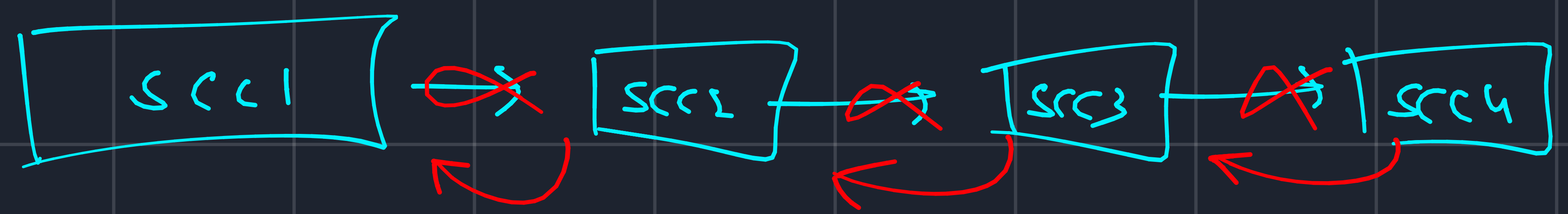


↳ steps

↳ Sort all the edges according to finishing time.

↳ Reverse the graph

↳ Do a DFS.



↳ But what can happen is 0 may be in s.c.c 4, then s.c.c 4 → scc 3

Due to reversal of edges.

↳ countered by starting & finishing time

→ sorting in finishing time -

DFS(0)

↓

DFS(1)

↓

DFS(2)

↓

DFS(3)

↓

DFS(4)

↓ ↑ ↘

DFS(7) DFS(5)

↓

DFS(6)

finishing order

stack

0

1

2

3

4

5

6

7

after reversal of edges.

↳ DFS(0)

DFS(2)

DFS(1)

→ SCC1

↳ DFS(3)

↳ DFS(7)

↳ DFS(4)

DFS(6)

DFS(5)

↳ 0, 2, 1

↳ 3

↳ 4, 6, 5

↳ 7